

Privacy Preserving Federated Intrusion Detection Framework

Real-Time IoT Network Security & Research Experiment Report

Generated: 2026-09-07 19:40:45 | **Framework:** PyTorch 2.9.1 / Python 3.10 | **Evaluation:** CICIDS2017 Held-Out Test Split

1. Executive Summary & Research Methodology

This project implements a decentralized intrusion detection architecture for Internet-of-Things (IoT) networks. Centralized baselines (Random Forest, XGBoost, and MLP) are evaluated independently on held-out test data. The federated system utilizes an MLP neural network whose trainable weight tensors are aggregated using sample-weighted Federated Averaging (FedAvg: $W = \sum (n_k / N) W_k$). **Academic Disclosure:** All mitigation mechanisms (Quarantine/Isolation) operate as software-level research simulations. Zero-sum masking simulates privacy-preserving aggregation properties; physical IoT battery/energy consumption is not claimed.

2. Model Performance Benchmark (Held-Out Test Set: 10,007 Samples)

Model	Accuracy	Precision (W)	Recall (W)	F1 Macro	F1 Weighted
Random Forest	99.80%	99.81%	99.80%	0.9881	0.9980
XGBoost	99.90%	99.90%	99.90%	0.9889	0.9990
Centralized Neural Network	98.91%	98.75%	98.91%	0.8029	0.9882
Federated Neural Network	97.01%	97.20%	97.01%	0.8404	0.9677

3. IoT Device Fleet & Dynamic Risk Status

Device Name	Type	IP Address	Attack Profile	Risk Score	State
Smart Camera	Camera	192.168.1.101	Normal + DDoS	87.3	SIMULATED ISOLATION
WiFi Router	Router	192.168.1.1	Normal + PortScan	100.0	SIMULATED ISOLATION
Temperature Sensor	Sensor	192.168.1.103	Normal + Botnet	80.0	SIMULATED QUARANTINE
Smart Door Lock	Door Lock	192.168.1.104	Normal + BruteForce	100.0	SIMULATED ISOLATION
Smart Light Controller	Smart Lighting	192.168.1.105	Normal + WebAttack	51.1	ALERT

4. Research Experiment History & Mathematical Verifications

Experiment	Mode	Protocol	Clients	Rounds	Accuracy	Macro F1	Comm (KB)
FL-NON_IID-FEDAVG-5R	NON_IID	FEDAVG	5	5	96.39%	0.6601	12739.0
FL-NON_IID-DROPOUT-5R	NON_IID	DROPOUT	5	5	97.21%	0.6883	11210.3
FL-IID-DROPOUT-10R	IID	DROPOUT	5	10	96.79%	0.6760	22420.6
FL-NON_IID-DROPOUT-5R	NON_IID	DROPOUT	5	5	96.77%	0.6289	11221.3
FL-IID-FEDAVG-3R	IID	FEDAVG	5	3	96.09%	0.6210	7650.9
FL-NON_IID-FEDAVG-5R	NON_IID	FEDAVG	5	5	96.73%	0.6277	12751.5

FL-NON_IID-SECURE_AGG-3R	NON_IID	SECURE_AGG	5	3	96.07%	0.6209	7650.9
FL-NON_IID-DROPOUT-5R	NON_IID	DROPOUT	5	5	96.79%	0.6293	11221.3
FL-NON_IID-FEDAVG-3R	NON_IID	FEDAVG	3	3	96.47%	0.6241	4590.5
FL-NON_IID-FEDAVG-3R	NON_IID	FEDAVG	5	3	96.09%	0.6210	7650.9
FL-NON_IID-FEDAVG-3R	NON_IID	FEDAVG	8	3	96.17%	0.6217	12241.4

5. Concept Drift 4-Stage Adaptation Telemetry

Drift Stage	Attacks Injected	Pre-Drift Acc	During Drift	Post-Adapt Acc
Stage 1: Volumetric Flooding	Normal, DDoS	97.1%	97.3%	98.2%
Stage 2: Endpoint Reconnaissance	DDoS, PortScan	97.1%	83.7%	79.2%
Stage 3: Credential & Web Exploitation	BruteForce, WebAttack	97.1%	34.5%	32.7%
Stage 4: Coordinated Botnet Sweep	DDoS, PortScan, BruteForce	97.1%	78.9%	73.0%

6. Telemetry Benchmarks & Explainability (XAI)

Measured Inference Latency: Mean = 0.43 ms | Median = 0.25 ms | P95 = 0.47 ms | Throughput = 2330.0 inferences/sec

XAI Verification: Fidelity Score = **93.3%** | Stability Score = **100.0%**

Academic Conclusion: The evaluated federated IDS framework successfully eliminates raw packet transfer while maintaining an accuracy within ~2.9% of centralized baselines under non-IID conditions. Real-time inference achieves sub-5ms latencies, demonstrating practical feasibility for edge IoT deployment.